Joint Variation Learning of Fusion and Difference Features for Change Detection in Remote Sensing Images

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***Abstract*— Remote sensing (RS) image change detection (CD) is an Earth observation technique for detecting surface changes in the same area during a period. With the rapid development of deep learning, various deep neural networks especially Siamese ones have been widely used in the field of CD. However, they have the deficiency of insufficient contextual information aggregation, thus resulting in false and missed detections, and it is difficult to refine the detection of change edges. To alleviate these problems and obtain more accurate results, we propose an efficient self-weighted spatial–temporal attention network (SSANet). In contrast to the Siamese structure, our network is a novel joint learning framework composed of fusion subnetwork, difference subnetwork, and decoder. Fusion subnetwork is used to extract multiscale object features where we propose a multicore channel-aligning attention (MCA) module to capture the long-range semantic information for multiscale context aggregation. Difference subnetwork is used to extract the difference variation features, where we propose a feature differential reconfiguration (FDR) module to learn the temporal change information. FDR can effectively filter change information and reconstruct features to improve the perception of changed regions. To better balance the MCA and FDR modules, an asymmetric weighting (AW) module is proposed in the decoder to self-weight the multiscale features and generate the change map. Experiments demonstrate the efficiency of proposed subnetworks and modules, and the state-of-the-art performance of SSANet.**

***Index Terms*— Asymmetric weighting (AW) module, change detection (CD), deep learning, feature differential reconfiguration (FDR) module, multicore channel-aligning attention (MCA) module.**

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I. Introduction

CHANGE detection (CD) is a task used to analyze the differences between bitemporal images in the same area at different times [1]. For a long time, CD in bitemporal remote sensing (RS) images has been one of the hot topics in the field of RS, which is used extensively for monitoring land resources [2], natural disasters [3], and urban extension surveys [4], [5]. As the progress of satellites and radar sensors in recent years, the temporal and spatial resolutions of bitemporal images have been gradually improved, which provides more complex information for CD task while also increases the difficulty of detection [6].

Traditional CD methods include principal component analysis (PCA) [7], change vector analysis (CVA) [8], and multivariate alteration detection (MAD) [9]. These methods accomplish the CD task mainly by analyzing the spectral information of the image and selecting the threshold value in the decision phase. However, these methods are more sensitive to disturbances, such as illumination variations, noise, and registration errors. Later, with the evolution of machine learning, learning-based methods such as support vector machines [10], decision trees [11], K-nearest neighbor (K-NN) [12], and random forest regression [13] were used in CD tasks. Machine learning algorithms can learn from partial samples to generate decision models to classify pixels into change and invariant classes. Since the features are manually extracted, they are difficult to represent high-level semantic features. Therefore, the accuracy of them has a lot of room to improve [14].

Over recent years, the rapid upsurge of big data and the improvement of hardware performance have created conditions for deep learning, and many deep learning-based neural network models have been applied in various fields, including the field of CD [15]. The convolutional neural network is a powerful deep training module that enables the automatic extraction of rich spatial–temporal features on diachronic images and maintains a good robustness to noise. Based on the excellent feature extraction capability of convolutional networks, many deep learning-based CD methods are proposed [16], [17].

Deep learning-based CD methods can be divided into patch-based and image-based models [18]. The patch-based approach uses the similarity of image patches as the basis for classification by judging them. This method is less efficient

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to compute because of the overlap between adjacent patches. Therefore, the mainstream deep learning-based methods are image-based methods because they enable image-to-image prediction in an end-to-end and efficient manner. In addition, the deep learning-based models are usually divided into early fusion (EF) and late-fusion (LF) methods due to the processing of bitemporal images [19]. EF method uses the concatenations or differences input of bitemporal images as model inputs. For instance, fully convolutional network (FCN) [20] is an end-to-end neural network model with a bitemporal image as input and output of the change map. After that, in order to improve the feature extraction ability, Ronneberger et al. [21] introduced the UNet, which uses skip connections to connect encoded and decoded streams. UNet has a stronger feature representation capability due to its use of feature concatenation in the skip connections phase. Later, the residual network [22] was also applied to the CD task for enhancing feature representation. However, as EF methods usually use single-branch networks, they are prone to information loss in bitemporal images during feature extraction. In contrast, the LF methods use separated bitemporal images as input, which greatly improves the accuracy due to the use of richer image features from the bitemporal images. For this reason, Siamese-based network is developed for CD as backbone, which consists of two or more identical subnetworks with shared weights [23]. For example, Daudt et al. [45] proposed a CD method by improving UNet with three models, including two LF models, which significantly enhances the detection capability of bitemporal images. Siamese-based network is usually used as encoders to extract features, and decoders are employed to synthesize features for prediction. Moreover, Jiang et al. [24] proposed the pyramid feature-based attention-guided Siamese network (PGA-SiamNet), which combined Siamese networks with various attentional mechanisms used to reinforce feature information. Wang et al. [25] proposed a metric-based Siamese network to obtain change maps by measuring the distance between features. Since the distance between unchanging features decreases and the distance of changing features expands, the metric-based approach can be used to enhance the discrimination between changed and unchanged regions, thus improving the detection accuracy.

Although deep learning-based methods achieve efficient detection performance, there still have some problems. First, as the resolution of RS images increases, the semantic information contained in the images becomes richer. It also contains more influencing factors such as image misalignment and illumination, and the current methods cannot aggregate the contextual information well [26]. Thus, more effective methods should be explored to extract more semantic information from images and to mitigate the impact of pseudo-change problem. Second, the current Siamese methods uses bitemporal images as input and extracts image features in the encoding stage, leading to a lack of focus on regions of change. This results in ambiguous predictions between changed and unchanged regions [27], [28]. Therefore, to enhance the ability of the network to extract the difference information, attempts to capture more discriminative information still need to be explored.

To address the above issues, this article proposes a novel joint framework and self-weighted spatial–temporal attention network (SSANet), which differs from the strategy of sharing weights adopted by the encoder in Siamese neural network architecture. It consists of an encoder and a decoder where the encoder is formed by two branches, a separate backbone fusion subnetwork, and an auxiliary differential subnetwork. We propose three modules to compose the network: multicore channel-aligning attention (MCA) module for the backbone fusion subnetwork, feature differential reconfiguration (FDR) module for auxiliary difference subnetwork, and asymmetric weighting (AW) module in the decoder stage.

In order to enhance the feature capture of bitemporal images and better learn the mapping relationship between images and ground truth, we use the fusion of bitemporal images as the input to the backbone fusion subnetwork and introduce the MCA module at this stage. The MCA module can extract multiscale features from fused images at different levels and obtain a more fine-grained representation, thus improving the integrity of object boundaries and the tightness of internal information in the features. It can better distinguish the object from the background. Moreover, in the CD task, since the changed area tends to be a smaller size than the unchanged region, the noise and illumination changes in practical images increase the difficulty in focus on the relevant information of the asynchronous image itself. The attention mechanism can discover the relationship between the spatial and temporal locations of the asynchronous image well, which makes it easier to distinguish the characteristics of different phases in a spatial and channel sense.

On the other hand, to enable the network to enhance the learning of changed regions and suppress irrelevant changes due to misalignment and noise, we propose the FDR module in the auxiliary difference subnetwork, which takes the difference of bitemporal images as input to learn and extract the intrinsic features of the real changing regions. We introduce deformable convolution in this module to locate and reconstruct the changing feature information, and the convolutional block attention module (CBAM) [29] is also introduced for learning more discriminative features.

Finally, in order to better use the fused features learned by the backbone fusion subnetwork and the difference features learned by the auxiliary difference subnetwork in the encoder stage, we design the AW module to perform upsampling, which can weight the channels in the feature maps during the upsampling process for extracting more robust information.

In summary, the main contributions in this article are summarized as follows.

1) We propose a novel end-to-end neural network architecture with an encoding phase that consists of two branches, a backbone fusion subnetwork paying attention to object features and an auxiliary difference subnetwork focusing on difference features, which is a new attempt in the field of CD to achieve integrated learning of CD tasks by fusing difference features.
2) An MCA module is designed to extract multichannel features in both short-range and long-range dimensions. It learns multiscale features independently and builds

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cross-channel interactions in a local way for refining semantic rich features.

3) An FDR module is introduced to reduce the influence of noise and uncorrelated changes on the detective results through reconstructing the difference map features and extracting the real change information.
4) An AW module is proposed to fuse object features and difference features using a channel weighting strategy, which enables the network to capture the contextual mapping relationships of multiple 2-D features in the decoding phase to generate change-aware representations.

The remainder of this article is presented as follows. Section II gives a reviewing of related work, Section III details the neural network approach and the various modules proposed in this article, and Section IV shows the experimental results and ablation experiments on different datasets. Finally, we discuss and summarize this article in Sections V and VI.

II. Related Work

In this section, we introduce related works, mainly including traditional RS image CD method and deep neural network models for CD.

A. Traditional CD Methods

In the early stages of CD, the detection task is usually performed with unsupervised methods, and the process is usually divided by three steps: 1) image preprocessing, such as image alignment and radiation correction [30]; 2) difference generation, where pixel-level differences are obtained by comparing features in dual-temporal images [31]; and 3) analysis of the difference map to generate CD results, usually obtained by thresholding segmentation or clustering algorithms [32]. Early threshold-based CD methods obtain features by algebraic operations, such as difference operator, algebraic ratio operator, and logarithmic ratio operator [33]. However, the results obtained by such methods depend severely on the choice of thresholds and experience of experts, so these methods are not suitable for handling complex scenarios. For this reason, clustering-based methods tend to be more popular in CD tasks.

Traditional clustering-based algorithms include c-means clustering and fuzzy c-means clustering (FCM) algorithms, which tend not to capture spatial information properly and are sensitive to the effect of noise [34]. Then, Gong et al. [35] proposed a fuzzy clustering algorithm based on the Markov random field (MRF) energy function. The algorithm achieved better performance, but the computational complexity also increased. To improve the detection efficiency, Ghosh et al. [34] proposed a fast FCM unsupervised CD algorithm. Moreover, Gong et al. [35] proposed a multiscale and multiresolution Gaussian mixture model. All these methods rely on difference map generation and have poor robustness to noise. In addition, some feature transformation techniques were used in CD tasks, such as PCA [7] and tasseled cap equivalent transformation [36]. The idea of these approaches is to classify the obtained difference images based on thresholding or clustering of pixels, finally obtaining change results by decomposing different change features at multiple scales [37]. It is worth mentioning that these methods rely heavily on the statistical properties of the image and are only suitable for extracting the spectral information of the image. Thus, they are more easy to ignore the background of bitemporal images [38].

B. Deep Learning-Based CD Methods

With the progress of deep learning, many neural network models and components are used in the field of CD to represent deeper level of feature information [39]. This makes it possible to detect changed region through end-to-end CD methods, and they have achieved good results in the field of CD. Convolutional neural networks are used to output more fine-grained change maps through patchwise category [40]. Mou et al. [41] proposed recursive convolutional neural networks which are capable to learn more spatial–temporal spectral information from multispectral images to generate better representations.

However, such networks fail to capture the contextual information properly because the patch size determines the convolutional field of perception. To address these problems, Long et al. [42] proposed an FCN, which was originally designed for use in the domain of image segmentation. The method achieves pixel-level output by replacing the fully connected layer with a fully convolutional (FC) layer, which avoids duplicate storage and computation due to the use of pixel blocks and is more efficient than recurrent neural networks. Therefore, FCNs are also widely used in CD tasks, and the network provides a new approach for CD tasks, which is based on the feature classification to get the final change map [43]. Subsequently, many different networks based on FCN have been generated, especially UNet [20], which innovatively introduces skip connections feature fusion approach to ensure that the network can extract finer features. Later, Peng et al. [44] proposed UNet++ network, which achieves multiscale feature extraction by using multiple layers of dense skip connections and suppresses pseudo-variation due to multiscale effects. However, the above single-branch networks do not make a good usage of bitemporal images, which results in information loss. Therefore, Siamese networks based on LF strategies are used in CD tasks and have become a mainstream framework with their efficient performance.

Siamese-based networks consist of two branches with shared parameters, it forms a new representation by mapping the input to the feature space, and then, the final difference result is obtained by the metric module. Daudt et al. [45] first proposed three different variants of the UNet-based network structure by exploring the training method for bitemporal image inputs, which are FC-EF based on image fusion input network structure, fully convolutional Siamese concatenation (FC-Siam-conc) network, and fully convolutional Siamese difference (FC-Siam-diff) network based on the Siamese-based structure. The experimental results show that Siamese-based networks have better performance, but they are still unable to extract multiscale feature map representations, making it difficult to satisfy CD tasks with complex change conditions. To exploit the full advantage of the temporal up-down relationship of bitemporal images, the

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spatial–temporal attention-based network (STANet) was proposed, which uses a Siamese-based structure and introduces a self-attention mechanism to model the spatial–temporal relationships [46]. Later, to focus on the difference features of bitemporal images, difference-enhancement dense-attention convolutional neural network (DDCNN) is proposed, which introduces a difference-enhancement (DE) module to weight each pixel for efficient feature aggregation [47]. In addition, Siamese-based network the combination of Siamese network and NestedUNet (SNU-Net) was proposed for the CD task [49], which enhances the representation of image features by the dense skip connections and ensemble channel attention modules. Meanwhile, in order to extract deep-level change information, Chen et al. [48] proposed a bitemporal image transformer network (BITNet), which introduces a bitemporal image converter to model remote spatial–temporal relationships in images. After the bit-temporal images are input to the network, the temporal features of the images should first be mapped to semantic tokens by convolution. The context is then modeled using a transformer-based encoder. The network innovatively applies the transformer to the CD tasks, which enables the network to focus on the long temporal semantic information and get better representation. However, the CD results are still affected by external forces, such as noise and illumination, which causes pseudo-change. To alleviate this problem, a deeply supervised (DS) attention metric-based network (DSAMNet) was presented [50], which captures more discriminative spatial–temporal features by introducing a CBAM [29] and deep supervised learning to achieve global feature extraction.

From the abovementioned methods, we can conclude that deep neural network models based on the Siamese framework mainly focus on improving model accuracy by enhancing feature model representation. However, since the input of Siamese network is bitemporal images, these networks are limited by the input images and cannot distinguish pseudo-change that is confusing in appearance, and cannot extract and fuse object features and difference features well, resulting in blurred changed regions and small region adhesion. The situation is serious, so the improvement that can be achieved by these methods is limited [51]. Therefore, in this article, we design a novel neural network framework for the CD task and new modules for capturing spatial and local information of diachronic images to achieve better performance and results.

III. Methodology

The current Siamese-based CD network has two identical and share-weight branches as encoders, and the inputs are bitemporal images at moments t1 and t2, respectively. It does not fully explore the inner connection of the bitemporal images. Fusion and difference of images can obtain rich object features and change features. However, networks that use image fusion independently as input often fail to focus on disparate features effectively, which leads to false detection. In addition, networks that use image difference as input independently often do not filter irrelevant change features well due to image misalignment and the presence of noise. Therefore, we propose to jointly input fusion and difference information in order to extract more discriminative features.

In this section, a novel CD network framework presented in this article is first introduced, after that the proposed network module is described in detail. The proposed network is shown in Fig. 1, where the network is structured in two phases, encoding and decoding stages, where the encoding phase is a joint learning network composed of two subnetworks, the backbone fusion subnetwork and the auxiliary difference subnetwork, respectively. Here, we propose the MCA module for composing the backbone fusion subnetwork and the FDR module for constructing the auxiliary difference subnetwork. The input of the fusion subnetwork is the contacted bitemporal images, and the difference subnetwork uses the difference of bitemporal images as the input. The difference map contains change information, which allows the subnetwork to strengthen its focus on differences and to reconstruct the difference features during the learning process. As a result, the responses of change and nonchange differ gradually during the learning process, which leads to a better definition of the changed region. In order to better fuse the global features and the fine-grained difference features at the object level, we propose the AW module for constructing the encoding stage. Finally, the model training is elaborated, including the training process and loss function.

A. Fusion Subnetwork

As shown in Fig. 1, the input of the fusion subnetwork is comprised of image fusion stitching of bitemporal images, followed by two 3×3 convolutional layers to achieve channel changes on the images. After that, the contacted images as input vectors X are downsampled by four MCA-based residual blocks. Four sets of features X_1, X_2, X_3, and X_4 are finally obtained.

1) MCA-Based Residual Block: Since the network encounters optimization and gradient disappearance problems as the neural network goes deeper, and the residual network has the advantages of alleviating degradation problems and promoting network convergence, in order to extract deep semantic features, we use the residual network as the feature capture module of the fusion subnetwork, and the designed MCA module is used instead of 3×3 convolution. Later, the CBAM module is added for feature aggregate, which can ensure that the semantic of the object will not be lost during the gradual downsampling [29]. As shown in Fig. 2, the residual module is formulated as follows:

$$X_{\text{out}}^{l} = \text{ReLU}\left(H\left(X_{\text{in}}^{l}\right) + \xi\left(X_{\text{in}}^{l}, \Omega_l\right)\right) \tag{1}$$

where X_{in}^{l} is the input feature map, and X_{out}^{l} is the output feature map after the MCA-based residual module. Separately, $H(\cdot)$ indicates identity mapping function, i.e., the right branch of the network, which consists of a 3×3 convolution. To keep the image size constant, the padding is set to 1, and the main purpose is to realize the conversion of the channels. $\xi(\cdot)$ denotes the residual mapping function, i.e., the left branch consisting of three convolutions, which are composed of 1×1 convolution, 3×3 convolution, and 1×1 convolution. The number of feature channels is gradually increased, and batch normalization is added at the end of each convolution in order to stabilize the convergence process. After that, the outputs

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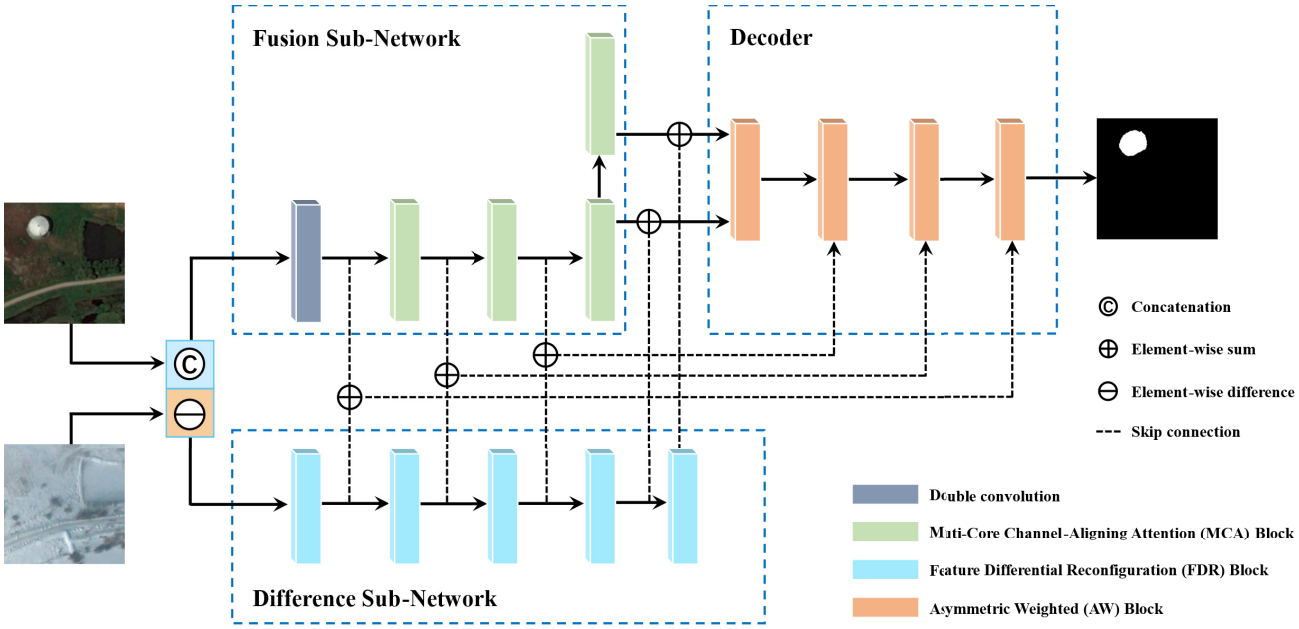


Fig. 1. Network structure.

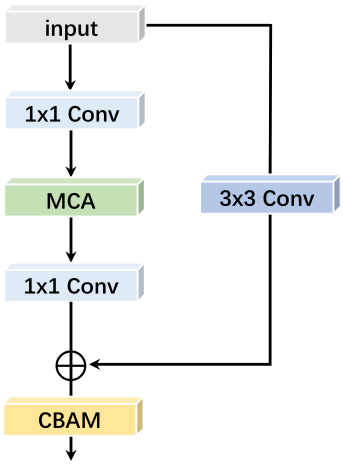


Fig. 2. MCA-based residual block.

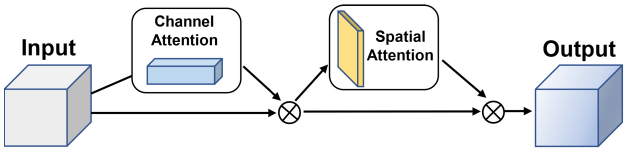


Fig. 3. Architecture of CBAM.

from $H(\cdot)$ and $\xi(\cdot)$ are summed and used as the input of the next layer, after which we add the Relu activation function for enhancing the learning and fitting ability of the network, which can be expressed by $f(z) = \max(0, z)$.

After that, the obtained features are reinforced by CBAM module so that the features can be further fused. As shown in Fig. 3, CBAM [29] is a lightweight module that contains two main parts, the channel attention module and the spatial attention module, which are used to enhance the discriminable information in different dimensions, where the channel attention module is intended to catch the contextual information in the channel direction, with the following equation:

$$M_c(F) = \sigma\left(\mathrm{MLP}^{\mathrm{r}}(\mathrm{AvgPool}(F)) + \mathrm{MLP}^{\mathrm{r}}(\mathrm{MaxPool}(F))\right) \quad (2)$$

where F represents the input feature of size $C \times H \times W$, and F generates two aggregated vectors of size $C \times 1 \times 1$ after max pooling and average pooling. After that, each vector is given a weight through a multilayer perception (MLP) module composed of two 1×1 convolutional layers. They are used for dimension reduction and dimension enhancement, respectively, and are connected by the Relu function in the middle. The resulting weights are multiplied with the original input features to obtain the interchannel rectified feature F'.

Analogous to the channel attention block, the spatial attention module also compresses the channels to get two vectors of size $1 \times H \times W$ after maximum pooling and average pooling, after which the final spatial attention map $M_s(F')$ is obtained after convolution layers and sigmoid function. This formula can be expressed as

$$M_s\left(F'\right) = \sigma\left(f^{(k\times k)}\left(\mathrm{AvgPool}\left(F'\right); \mathrm{MaxPool}\left(F'\right)\right)\right). \quad (3)$$

Then, the final feature map F'' is obtained, and the whole process is as follows:

$$\begin{aligned} F' &= M_c(F) \otimes F \\ F'' &= M_s(F') \otimes F'. \end{aligned} \quad (4)$$

In contrast to the original residual network, the fusion subnetwork has many downsampling operations, thus causing the loss of details in deep features. Even if a CBAM module is added after each residual module to alleviate this type of problem, it is still difficult to maintain the integrity of

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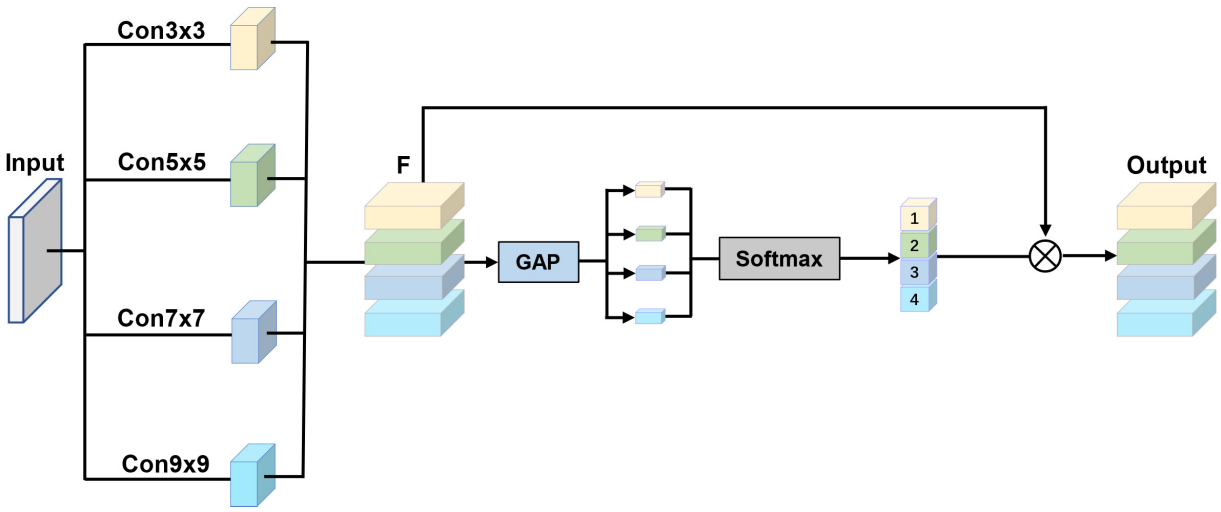


Fig. 4. MCA module.

the objects in deep features, and it is difficult for a fixed convolution to capture the global features of different objects. Therefore, to alleviate this problem, we design an adjustable multiscale convolution module that can select different convolutions according to the object size, and thus capture different contextual information.

2) MCA Module: As shown in Fig. 4, in the MCA module, we introduce the mechanism of multicore convolution, where first the input feature X is divided by channel to four parts, such as $X_i[X_1, X_2, X_3, X_4]$. For each divided feature map X_i, the size is $(C/S) \times H \times W$. It is notable that the number of channels C is divided by S which is set to 4 in this article. The feature maps are divided equally based on the number of channels. The purpose of doing so is to process the features at multiple scales so as to obtain resolution and scale information in different spaces. After that, we introduce the method of group convolution and send each group of feature maps into the convolution with the convolution kernel size of 3, 5, 7, and 9, respectively, and the stride is 1, in order to ensure that the output feature size remains unchanged. We set the padding of each convolution kernel to 1, 2, 3, and 4. The group convolution formula is as follows:

$$F_i = \text{Conv}(k_i \times k_i)(X_i) \quad i = 0, 1, 2, \ldots, S-1 \tag{5}$$

where i represents the ith convolution, and the corresponding convolution kernel size is $k_i = 2 \times (i+1) + 1$. X_i represents the corresponding feature map after division, and F_i represents the feature map after convolution of different scales. Finally, we splice four sets of different features to get the next preprocessing feature map, which can be expressed as

$$F = \text{Cat}\left(\left[F_0, F_1, \ldots, F_{S-1}\right]\right) \tag{6}$$

where Cat stands for channel concatenation operation, F is the set of multiscale feature maps, and then, we extract the attention weight information based on the channel to obtain weight vectors of different scales. First, we compress the global information to the channel dimension by global average pooling. To better utilize the compressed information in the channel, we use two fully connected layers to parameterize the channel information, and then use the sigmoid activation function to map the result to a controllable interval. The formula is as follows:

$$S_i = \sigma(g(\mathbf{z})) = \sigma\left(\text{FC}_{1,2}(A(F_i))\right), \quad i = 0, \ldots, s-1 \tag{7}$$

where S_i is the attention weight from F_i and $A(\cdot)$ is the global pooling operation, the purpose is to compress the global information to the channel dimension, and $\text{FC}_{1,2}(\cdot)$ is the two fully connected layers, which can better utilize the compressed global information by parameterizing the channel information. Then, map the result to a controllable interval through the activation function $\sigma(\cdot)$. Then, the softmax function is performed based on S_i to adaptively select different spatial scales. The weight allocation formula is as follows:

$$\text{att}_i = \text{Softmax}(Z_i) = \frac{\exp(S_i)}{\sum_{i=0}^{S-1} \exp(S_i)}. \tag{8}$$

The softmax function is used to obtain the multiscale calibration weight att_i, which contains channel weights and compressed position information. Through this operation, the interactive adjustment of local and global attention can be achieved, and then, the multiscale channel weights are multiplied by the relevant scale feature maps F_i as follows:

$$Y_i = F_i \odot \text{att}_i \quad i = 1, 2, 3, \ldots, S-1 \tag{9}$$

where $\odot$ represents the multiplication of the channel dimensions, Y_i is the feature map after multiscale attention weight correction, and finally, we splice the obtained multiscale feature maps to obtain a weighted feature vector F'' with the identical size as the input feature. The formula is as follows:

$$F'' = \text{Cat}\left(\left[Y_0, Y_1, \ldots, Y_{S-1}\right]\right). \tag{10}$$

As shown in the above analysis, the proposed MCA module can combine multichannel spatial information and cross-channel attention mechanism into each segmentation feature group. The interaction of local and global channel attention is achieved, which can capture more complete characterization information.

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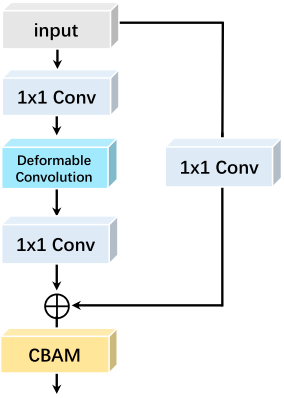


Fig. 5. FDR module.

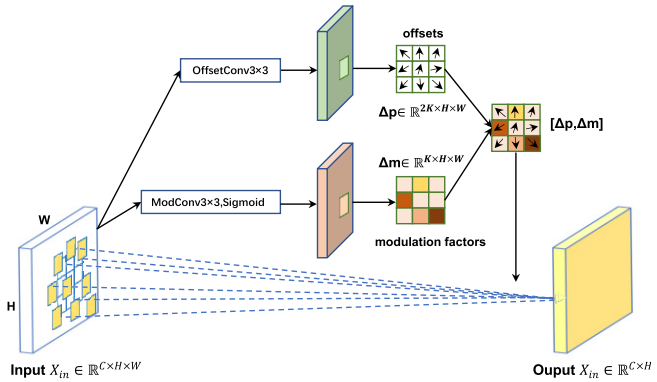


Fig. 6. Deformable convolution.

B. Difference Subnetwork

As shown in Fig. 1, the input of the difference subnetwork is the difference of bitemporal images, which contains five FDR modules to perform feature extraction on the input image. The obtained feature maps of different stages are summed with the feature maps generated by the fusion subnetwork in the decoder stage Thus, the representation of change information is enhanced.

1) FDR Module: The FDR module is presented in Fig. 5, and its network structure is analogous to the MCA-based residual block. With the difference, in the right branch of the network, we use a 1×1 convolution instead of the 3×3 convolution of the MCA-based residual block. To endow the network with different perceptual domains to efficiently capture and reconstruct the disparity information, we exploit a deformable convolution in the residual module [52].

For normal 2-D convolution, the convolution process can be defined into two steps: 1) sampling the input feature X using a convolution kernel and 2) the weighted sum of samples is obtained as the output of this convolution, e.g., a convolution kernel of size 3×3 with position $p_i \in \{(-1,-1),(-1,0),\ldots,(1,1)|i = 9\}$. For each point p_i in each output feature map, the convolution formula is as follows:

$$y(p_0) = \sum_{p_n \in R} w(p_n) \cdot x(p_0 + p_n) \tag{11}$$

where $w(p_n)$ denotes the weight of the corresponding position of the convolution kernel. Due to the fixed structure of convolution neural network, the convolution kernel can only sample the input features at a fixed position, and the fixed perceptual field is defective for capturing different spatial information. While the input disparity map is obtained by making differences in the dual-temporal images, so it will be affected by noise and illumination and make the disparity features difficult to focus on. Therefore, we introduce feasible variable convolution in the process of extracting disparity variation, using self-adaptive attention to useful feature information in the disparity input, which suppresses irrelevant noise and is used to assist the MCA-based fusion subnetwork to generate features with more scale-rich and semantic information.

The deformable convolution achieves the adjustment of the perceptual field by adjusting the offset factor and modulation factor of the sampling position, and its convolution equation is as follows:

$$X_{\text{out}}(p) = \sum_{k=1}^{K} w_k \cdot X_{\text{in}}(p + p_k + \Delta p_k) \cdot \Delta m_k \tag{12}$$

where Δp_k and Δm_k denote the 2-D offset and modulation factor at the Kth position in 2-D convolution, respectively, where the 2-D offset mainly includes the offset in the x- and y-directions.

As shown in Fig. 6, let X_{in} denotes the input feature map, p_i denotes the center position, and the deformable convolution is mainly implemented by two convolution layers, one is a 3×3 convolution called OffsetConv to get the offset Δp_k and another is a 3×3 convolution called ModConv to obtain the modulation factor Δm_k. As mentioned above, the 2-D offset includes the offset in both x- and y-directions, so the number of convolution kernel filters of OffsetConv is set to $2K$, which is used to generate the 2-D offset of the feature maps, i.e., $\Delta p = \{\Delta p_k^i\}_{k=1}^K = \{\Delta x_k^i, \Delta y_k^i\}_{k=1}^K \in \mathbb{R}^{2K \times H \times W}, 1 \le i \le H \cdot W, K = 9$. ModConv denotes the weight size of each sample point, so the number of filters is set to K. The modulation factor is generated by the sigmoid activation function, i.e., $\Delta m = \{\Delta m_k^i\}_{k=1}^K \in \mathbb{R}^{K \times H \times W}, 1 \le i \le H \cdot W, K = 9$. The sigmoid activation function allows the modulation factor to be mapped to the range(0, 1), which can be expressed as $f(z) = (1/1 + \exp(-z))$.

To implement the backward propagation of the feasible variable convolution, we use bilinear interpolation to obtain the pixel values of $X_{\text{in}}(p + p_k + \Delta p_k)$. In summary, the deformable convolution process is as follows. First, it learns the offsets $\{\Delta p_k = (\Delta x_k, \Delta y_k)\}_{k=1}^K$ in the x- and y-directions and the modulation factor $\{\Delta m_k\}_{k=1}^K \in (0, 1)$ at each sample point based on the input features, the purpose of offset is to find the location of the region of the target information, and the modulation factor is to assign a weight to the location found. These two aspects ensure the accurate extraction of the deformable convolution. After that, we use the adjusted feature map as a new input and generate the output feature map after a normal convolution.

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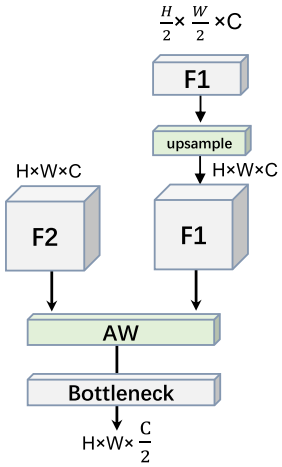


Fig. 7. AW-based decoder block.

C. Decoder

In the proposed network, the encoder generates low-level semantic features, which contain accurate location and edge information. The decoder generates high-level semantic features, which have more semantic detail content. In the network decoding process, the low-level and high-level feature maps need to be stitched together and later go through the upsampling module to generate the disparity map. When the features go through the decoder, the ordinary decoder has limited extraction between shallow and deep features because it often just concatenates low-level feature maps and high-level feature maps. In order to explore the feature reconstruction capability of the network, we propose an AW decoding module, which can effectively balance the relationship between low-level and high-level feature maps, thus allowing different features to supervise and complement each other.

The decoder block structure is shown in Fig. 7, and the input of this network comprises of the feature maps F_i generated in the encoder stage, in which we upsample the smaller feature maps because of the inconsistent size of F_i images. After that for the two feature maps of the identical size enter the AW module to get the weight sizes of different features, the conversion between channel dimensions is performed by a bottleneck residual block, thus realizing the process of a decoding module.

1) Asymmetric Weighted Module: To enable the decoding module to self-balance the feature relationships, we introduce an AW module with the structure shown in Fig. 8, which can selectively weight between different feature maps. Specifically, the first input is the two branches of feature information from the encoder and decoder, and in order to adjust the weighting information of different features, there is a necessity to integrate the features of all branches, first by summing the feature information of the two branches together

$$F = F_1 + F_2. \tag{13}$$

After that we embed the global information by global pooling to generate the channel-dependent feature vector s_c,

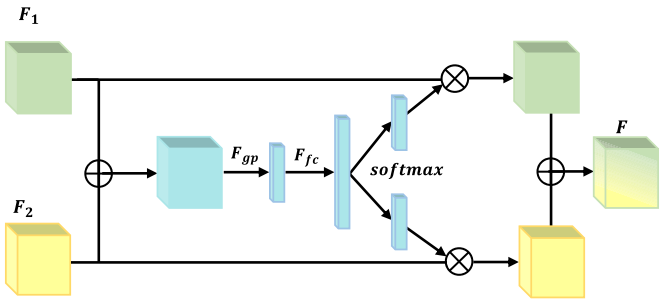


Fig. 8. AW module.

i.e., which is obtained by compressing the spatial dimension of F

$$s_c = \mathcal{F}_{\text{gp}}(\mathcal{F}) = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} F(i, j). \tag{14}$$

After the batch normalization and Relu activation functions to normalize the data to a normal distribution, the channel dimensionality reduction is performed through a fully connected layer

$$z = \mathcal{F}_{\text{fc}}(s) = \delta(B(s_c)) \tag{15}$$

where $\mathcal{B}(\cdot)$ is the regularization and $\delta(\cdot)$ denotes the Relu activation function. After that z goes through softmax to generate two vectors a and b, using the weight information from the adaptive selection of different features, where $a + b = 1$

$$F^{'} = F_1 \times a + F_2 \times b. \tag{16}$$

By the above operation, F' contains semantic-rich image features of both low-level and high-level feature maps after weighting, thus providing more useful information for the CD task.

D. Network Details

1) Training Process: Training procedure of the network consists of an encoder and a decoder. The coding part contains two parts, the fusion subnetwork and the difference subnetwork; the fusion subnetwork contains a general convolutional module and four MCA modules, and the difference subnetwork contains a general convolutional module and four FDR modules, so both subnetworks generate five feature maps in the coding stage. We let the feature maps corresponding to each stage be summed to obtain the joint features of each stage, with each stage corresponding to the number of channels 64, 128, 256, 512, and 512. The feature map size ($H \times W$) is halved from 256 × 256 to 16 × 16 in that order. In the decoder, which includes four AW blocks and a 1 × 1 convolution used for classification, so four skip connections are used to implement the upsampling of the network features and a classifier is joined in the last layer to produce the difference map, channels corresponding to each stage are 256, 128, 64, 64, and 1. The feature map sizes ($H \times W$) are 32 × 32, 64 × 64, 128 × 128, 256 × 256, and 256 × 256, respectively.

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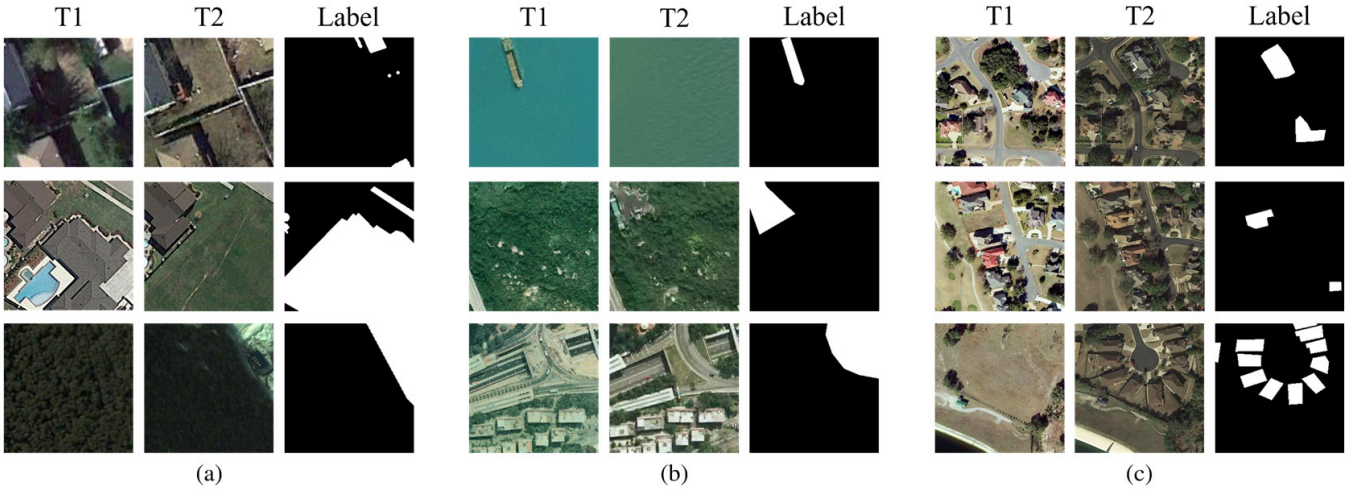


Fig. 9. Exhibition of some samples in different datasets. (a) CDD (b) SYSU-CD (c) LEVIR-CD.

2) Loss Function: For the CD task, the samples that changed regions in an image are often smaller than the samples that unchanged regions, which makes the network prone to bias during training, thus resulting in the effect of loss imbalance. To reduce this, we use the batch balance contrast loss as the loss function with the aim of modifying the weights of different loss terms of the loss function to achieve a balancing effect. The loss function formula is as follows:

$$L\left(D^*, M^*\right) = \frac{1}{2}\frac{1}{p_u}\sum_{b,i,j}\left(1 - M^*_{b,i,j}\right)D^*_{b,i,j} + \frac{1}{2}\frac{1}{p_c}\sum_{b,i,j} M^*_{b,i,j}\mathrm{Max}\left(0, m - D^*_{b,i,j}\right) \tag{17}$$

where D^* denotes the difference map obtained by the network, M^* denotes the binary label, 1 denotes the changed value, and 0 denotes the unchanged value. b, i, and j denote the batch, height, and width, respectively. m denotes the boundary value limiting the changed pixels, which is set to 2 in the experiment. p_{u} indicates the number of unchanged pixel pairs, and p_{c} represents the number of changed pixel pairs. They are calculated based on the label of the corresponding category

$$p_{\mathrm{u}} = \sum_{b,i,j} 1 - M^*_{b,i,j},\ p_c = \sum_{b,i,j} M^*_{b,i,j}. \tag{18}$$

IV. Experiments

In this section, experiments are carried out on three public CD datasets, CDD [53], Sun Yat-Sen University change detection dataset (SYSU-CD) [50], and LEVIR building change detection dataset (LEVIR-CD) [46]. To demonstrate the effectiveness of the proposed methods in this article, we experimentally compare several state-of-the-art deep learning-based methods, which are FC-EF [45], FC-Siam-conc [45], FC-Siam-diff [45], STANet [46], BITNet [48], SNU-Net [49], DSAMNet [50], and DDCNN [47]. After that, common evaluation metrics are calculated and these methods are compared and analyzed. Finally, ablation experiments are used to verify the efficiency of the proposed modules and use visualization of the metrics for further analysis.

A. Dataset Description

1) CDD Dataset: This dataset is a seasonal variation dataset, 0.03–1 m per pixel, including seven pairs of high resolution (HR) images of size 4725 × 2200 pixels and four pairs of images of size 1900 × 1000 pixels. After cropping and rotation, 16 000 pairs of seasonally changing Google Earth images with pixel change labels are obtained, and each RS image is 256 × 256 pixels of size. Among them, 10 000 images were used for net training, 3000 images for validation, and 3000 images for testing. As shown in Fig. 9(a), the CDD dataset not only contains common change information such as buildings and trees, but also provides change information of lots of detailed objects such as roads and cars. In order to prevent overfitting and speed up convergence, we perform necessary data augmentation operations, including random rotating and flipping.

2) SYSU-CD Dataset: The SYSU-CD dataset contains information on multiple types of changes in complex scenes, encompassing different types of changes in Hong Kong's high-rise buildings, marine ports, vegetation changes, and suburban sprawl. As shown in Fig. 9(b), after data enhancement, SYSU-CD dataset contains 20 000 pairs of aerial images with a size of 256 × 256 with 0.5-m resolution. In the experiments, the ratio of training, validation, and testing is divided into 6:2:2.

3) LEVIR-CD Dataset: The LEVIR-CD dataset contains 637 pairs of VHR Google Earth images with a size of 1024 × 1024 and resolution of 0.5 m. The dataset focuses on changes in building examples and covers 20 different regions in TX, USA. As shown in Fig. 9(c), it has a total of 31 333 individual variations of buildings, covering villas, apartments, garages, shopping malls, and other types of buildings. In our experiment, the ratio of training, validation, and testing is divided into 7:1:2. Each image is cropped to 256 × 256 size. To prevent overfitting, we perform the necessary data enhancement operations, including random rotating and flipping.

B. Compared Methods

To validate the superiority of the proposed approach, eight state-of-the-art CD methods have been selected for comparative testing and are sketchily stated as follows.

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1) FC-EF [45]: It is based on the fused input UNet network, where the bitemporal RS images are combined with the decoder through jump connections after entering the encoder to generate disparity information.
2) FC-Siam-conc [45]: The network is a variant of the FC-EF network, where the encoder is a Siamese network with shared weight, and the net links evenly the three feature maps of the encoder branches and decoder for jumping to obtain the change information.
3) FC-Siam-diff [45]: As another variant of FC-EF, the absolute value of the difference between the feature maps from the two decoders is first obtained, and then, a skip connection is made with the corresponding layer of the decoder.
4) STANet [46]: A metric-based Siamese network, which introduces a spatial–temporal attention mechanism to enhance features.
5) BITNet [48]: A converter-based approach to contextual modeling by semantic tagging of input images, followed by reduction of features by a conversion decoder.
6) Combination of SNUNet [49]: A multiscale stitching approach applies the Siamese network structure to Nest-UNet, incorporating channel attention at each layer, and deeply supervised methods are used to enhance feature discrimination.
7) DSAMNet [50]: A network based on Siamese structure, where the feature extractor is used to get multiscale features, and then, a metric module is used to learn distance mapping, and a two-layer deep supervision module is used to increase the feature extraction capacity to produce more helpful features.
8) DDCNN [47]: A network based on UNet structure, which introduces a dense attention method to model the correlation between deep features and shallow features. In addition, the network introduces a DE module to selectively aggregate features.

C. Experimental Setup

1) Evaluation Metrics: To validate the performance of the proposed network, we use four indicators to assess the similarity between the predicted results and the true variation, including precision (Pre), recall (Rec), F1-score (F1), and intersection over union (IoU), where Pre represents the false positive rate, and Rec reflects the leakage rate. F1 is the harmonic mean of Pre and Rec, and IoU is the ratio of the overlap area of the prediction result and the true label to its union. The formula for each indicator is as follows:

$$\text{Pre} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{19}$$

$$\text{Rec} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{20}$$

$$\text{F1} = \frac{2\text{Pre} \cdot \text{Rec}}{\text{Pre} + \text{Rec}} \tag{21}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{22}$$

TABLE I
COMPARATIVE EXPERIMENTAL RESULTS ON CDD DATASET

Methods	Pre (%)	Rec(%)	F1(%)	IoU(%)
FC-EF	84.99	64.14	73.11	57.62
FC-Siam-diff	82.01	83.89	82.94	70.85
FC-Siam-conc	86.66	81.73	84.12	72.60
STANet	89.42	93.70	91.51	84.35
BITNet	91.11	**98.06**	94.45	89.50
SNUNet	94.83	94.16	94.49	89.57
DSAMNet	95.96	91.77	93.81	88.35
DDCNN	97.63	91.24	94.33	90.65
Ours	**99.07**	96.06	**97.54**	**95.20**

where TN and TP indicate the number of true negatives and true positives, respectively, and FN and FP indicate the false negatives and false positives.

2) Implementation Details: Our proposed network model is structured in the pytorch architecture, which is trained and tested on an NVIDIA GeForce GTX3080 GPU with 12G of graphics memory. As with other comparative experiments, we implement a common data enhancement process such as flip and Gaussian blur on the input images. Adam gradient descent is utilized to optimize the model. During the training period, the batchsize, learning rate, and momentum are set to 8, 0.000125, and 0.99, respectively.

D. Comparing Experiments

1) Comparisons on CDD Dataset: The evaluation metrics of the CDD dataset are shown in Table I, where FC-EF has the lowest F1 and IoU among all compared methods, followed by the FC-Siam-diff network with F1 and IoU of 82.94% and 70.85%, respectively. This shows that the Siam network can extract the change information more effectively than the unilateral network. After that, FC-Siam-conc scores 86.66% on Pre, and its F1 and IoU are 84.12% and 72.60%, respectively, which means that the network can retain more useful information by feature splicing than making differences in the decoder stage. The F1 and IoU of STANet are 91.51% and 84.35%, respectively, which indicates that the attention mechanism can effectively reinforce the feature information. The metrics of BITNet are higher compared to the above networks, with F1 and IoU reaching 94.45% and 89.50%, respectively, indicating that using transformer-based modeling of context can effectively enhance the change information for higher accuracy. The network of SNU-Net performs a little better than BITNet, with F1 and IoU at 94.49% and 89.57%, respectively. This shows that intensive linking works well for feature aggregation. The

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DSAMNet performs similar to BITNet and SNU-Net, but its Pre is 95.96%, which is significantly higher than that of BITNet and SNU-Net, which indicates that the depth-based supervised strategy can predict the better change categories well. Then, the F1 and IoU of DDCNN are 94.33% and 90.65%, respectively, which indicates that the DE module can effectively distinguish the changed regions. Finally, among all the methods of comparison, our proposed method achieves the highest value in most metric, with IoU at 95.20%, F1 at 97.54%, Rec at 96.06%, and Pre at 99.07%, which is 4.55%, 3.21%, 4.82%, and 1.44% higher than DDCNN.

Fig. 10 provides graphs of the test results of different methods on the CDD dataset. As can be seen, the first three methods FC-EF, FC-Siam-diff, and FC-Siam-conc can only identify relatively obvious changes in the images, such as building changes and road changes. However, they are unable to detect changes in small objects, such as cars and trees, which are missed in the detection and result in the relatively low accuracy rate. STANet has fewer missed detections, but it still has more false detections. The transformer-based BITNet can capture small object changes, but BITNet does not keep the boundaries of small objects well and is prone to false detection. Although the SNUNet reduces false detections, it also leads to relatively coarse and incomplete detection results compared to the true values. Compared with the SNUNet, DSAMNet performs well for cars, but still does not capture building changes very well. After that DDCNN achieves better results, but the degradation of contextual information in the feature decoding stage leads to the inability to discriminate pseudo-variations, as shown in the first row in Fig. 10. Because the road in Fig. 10 has a similar color behavior to the background, it can be seen that these methods do not identify these variations. As seen in the graphic, our proposed method can well fuse the difference information and feature information, and weight different information adaptively to obtain higher accuracy discriminative features.

2) Comparisons on SYSU-CD Dataset: Table II shows the test results of different methods on SYSU-CD dataset; since the SYSU-CD contains relatively complex change information, it can be seen from the results that our network also performs better in dealing with complex and changing datasets. The individual evaluation metrics of the proposed network are significantly superior to the above methods, with IoU and F1 being 68.18% and 81.08%, respectively, followed by DDCNN with IoU and F1 of 65.34% and 79.12%, respectively, which demonstrates the effectiveness of the DE module. Later, the DSAMNet again proves the effectiveness of deep supervision with IoU and F1 of 64.25% and 78.23%, respectively. The IoU and F1 of the SNUNet were 64.02% and 78.06%, respectively. The transformer-based BITNet also achieves a good performance with 88.45% Rec, which shows that the BITNet pays better attention to contextual information and achieves fewer missed detections. The STANet is similar to the BITNet with IoU and F1 of 77.27% and 63.05%, respectively. After that, the performance decreases in the order of FC-Siam-conc, FC-EF, and FC-Siam-diff. Among the three networks based on the UNet and Siamese framework, FC-Siam-conc has the best performance with 62.53% and 76.95% for IoU and F1, respectively, which indicates the good feature extraction performance of Siamese fusion network. It is worth noting that FC-EF performs better than the FC-Siam-diff network in the SYSU-CD, and the IoU and F1 of FC-EF are scored as 60.18% and 75.14%, respectively. Although the Pre of the FC-Siam-diff network is higher, but the SYSU-CD dataset has a relatively complex scene, the method of using features to do the difference directly will lead to network overfitting and false detection. Therefore, its F1 and IoU are the lowest.

TABLE II
COMPARATIVE EXPERIMENTAL RESULTS ON SYSU-CD DATASET

Methods	Pre(%)	Rec(%)	F1(%)	IoU(%)
FC-EF	81.61	69.63	75.14	60.18
FC-Siam-diff	63.58	87.08	73.50	58.11
FC-Siam-conc	82.36	72.20	76.95	62.53
STANet	75.40	79.63	77.27	63.05
BITNet	68.78	**88.45**	77.39	63.12
SNUNet	77.45	78.68	78.06	64.02
DSAMNet	78.80	77.68	78.23	64.25
DDCNN	79.71	78.54	79.12	65.34
Ours	**82.48**	79.73	**81.08**	**68.18**

Fig. 11 shows the results attained by different methods on the SYSU-CD dataset, which contains complex change objects such as roads, ships, and buildings. We can see that the recognition of buildings by FC-EF is not accurate enough. There exist many false detections in the detection results of FC-Siam-diff, which also confirms the low Rec in Table II. FC-Siam-conc can identify some major changes, but there are still more false detections. The STANet with the attention mechanism can further identify regions of change, but there are many missing detections. As BITNet introduced the transformer mechanism, it can get more realistic changes, but the performance in complex scenes is poor. In most cases, SNUNet and DSAMNet can capture a more complete change area, including changes in roads and ships. However, it cannot identify changes with small differences in appearance properly. The performance of DDCNN has further improved, but it still has more significant missed detections, such as the last row in Fig. 11. In conclusion, our proposed method enables to extract better integrated feature maps of changed regions. Hence, we get fewer false and missed checks in the detection area. Similar to the results on the CDD dataset, it has a better representation in maintaining the change boundary information.

3) Comparisons on LEVIR-CD Dataset: Table III ws the test results of different methods on LEVIR-CD dataset; since

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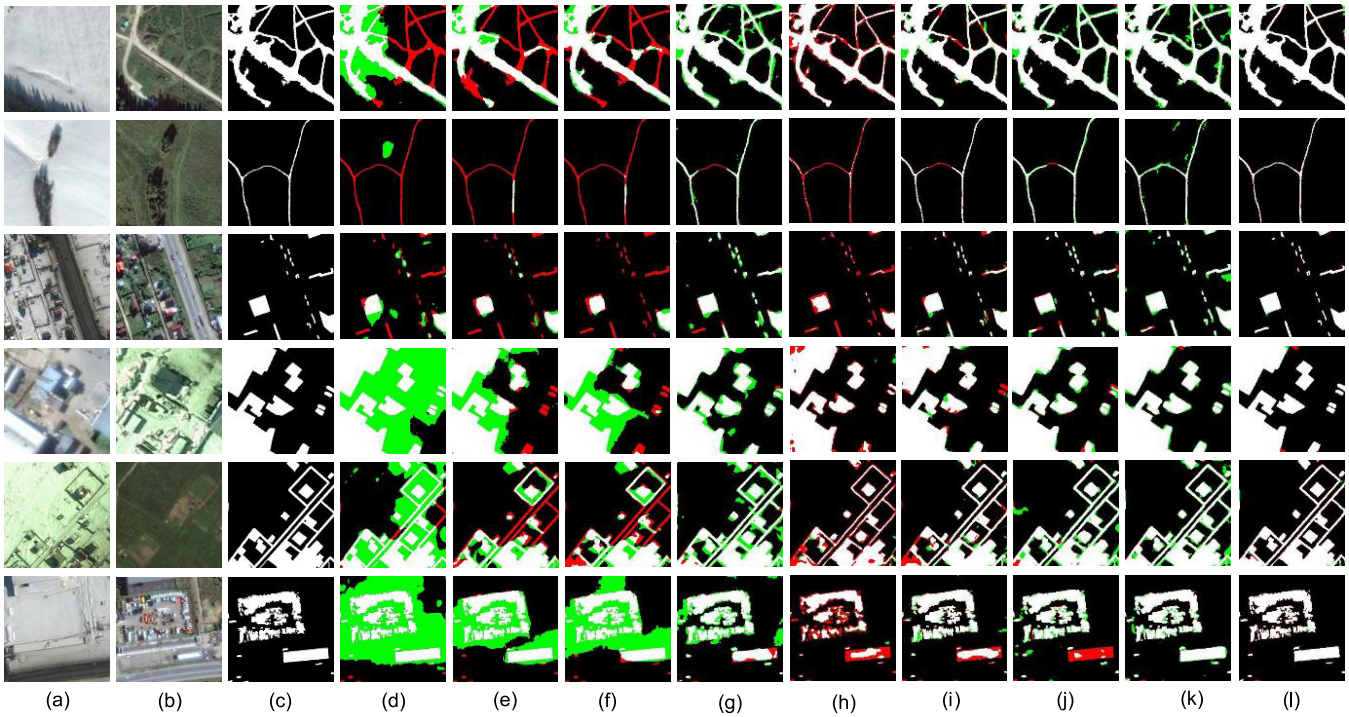


Fig. 10. Visual comparisons on the CDD dataset: (a) image at time 1; (b) image at time 2; (c) ground truth; (d) FC-EF; (e) FC-Siam-diff; (f) FC-Siam-conc; (g) STANet; (h) BITNet; (i) SNUNet; (j) DSMANet; (k) DDCNN; and (l) proposed; colors: white for TP (i.e., "changed"), black for TN (i.e., "unchanged"), red for FP, and green for FN.

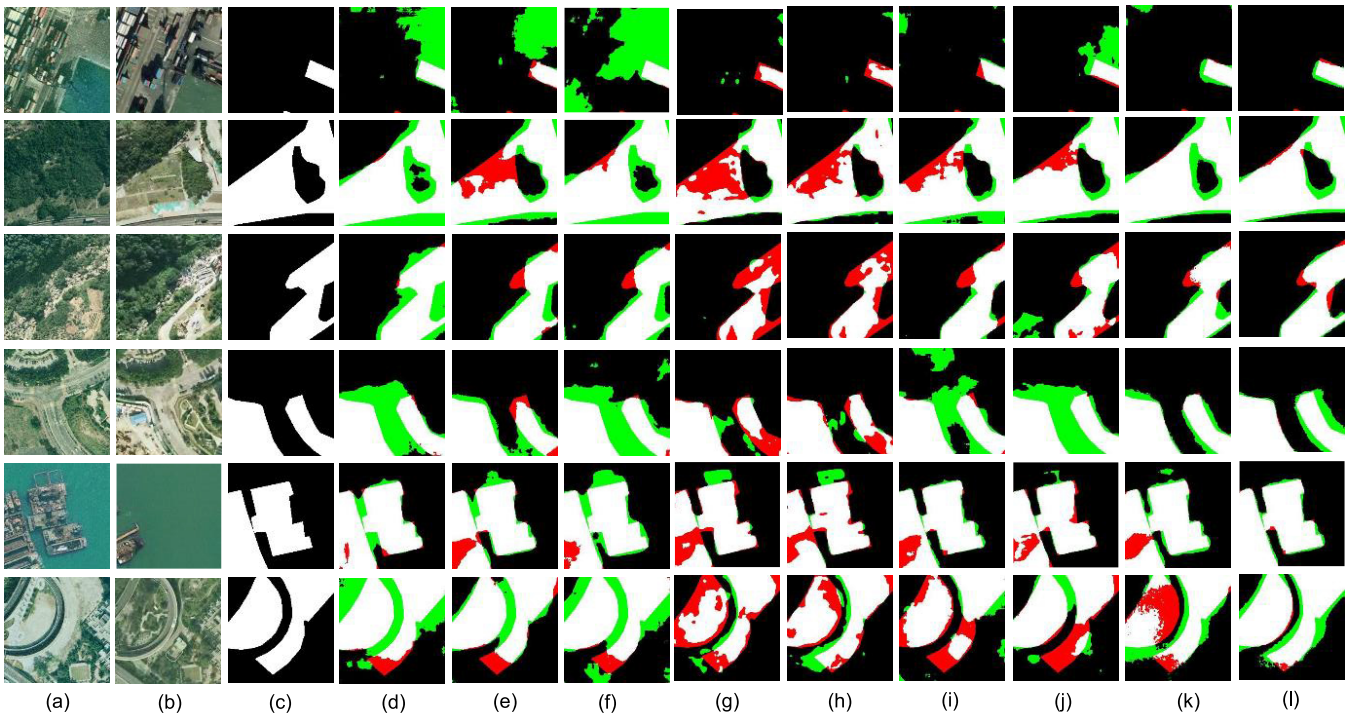


Fig. 11. Visual comparisons on the SYSU-CD dataset: (a) image at time 1; (b) image at time 2; (c) ground truth; (d) FC-EF; (e) FC-Siam-diff; (f) FC-Siam-conc; (g) STANet; (h) BITNet; (i) SNUNet; (j) DSMANet; (k) DDCNN; and (l) proposed; colors: white for TP (i.e., "changed"), black for TN (i.e., "unchanged"), red for FP, and green for FN.

the LEVIR-CD contains relatively complex change information, it can be seen from the results that our network also performs better in dealing with complex and changing datasets. The individual evaluation metrics of the proposed network are significantly superior to the above methods, with IoU and F1 being 83.69% and 91.12%, respectively, followed by

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TABLE III
COMPARATIVE EXPERIMENTAL RESULTS ON LEVIR-CD DATASET

Methods	Pre(%)	Rec(%)	F1(%)	IoU(%)
FC-EF	84.67	81.03	83.68	71.94
FC-Siam-diff	89.11	83.02	85.96	75.37
FC-Siam-conc	92.64	76.89	84.03	72.46
STANet	87.99	87.65	87.82	77.51
BITNet	89.93	89.45	89.69	81.31
SNUNet	89.14	86.73	87.92	78.45
DSAMNet	80.19	89.06	84.39	73.01
DDCNN	**93.16**	87.34	90.16	82.08
Ours	91.71	**90.53**	**91.12**	**83.69**

DDCNN, which achieves IoU and F1 of 82.08% and 90.16%, respectively, proving the efficiency of its DE module. After that, it is the transformer-based BITNet, which shows better results on the LEVIR-CD dataset with IoU and F1 of 81.31% and 89.69%, respectively. The SNUNet shows IoU and F1 as 78.45% and 87.92%, respectively, which demonstrates that dense skip connections can effectively aggregate contextual features. The STANet based on the attention mechanism has similar results to SNUNet with IoU and F1 of 77.51% and 87.82%, respectively. It is worth noting that DSAMNet cannot achieve better results on the LEVIR-CD dataset, which has 73.01% and 84.39% for IoU and F1, respectively. Among the three networks based on the UNet and Siamese framework, FC-Siam-diff has the best performance with 75.37% and 85.96% for IoU and F1, respectively.

Fig. 12 shows the results attained by different methods on the LEVIR-CD dataset. We can see that three networks based on UNet and Siamese framework are not accurate enough for building identification. The DSAMNet suffers from the same problem, with significant false detections and missed detections in its results. SNUNet and STANet can capture more complete regions, but there are still more false detections. BITNet and DDCNN can capture features more accurately, with fewer missed and false detections, but they still cannot identify more complex changes. As can be seen in Fig. 12, our proposed method can better extract the integrated feature maps of the changed regions. We get fewer false and missed checks in the detection region. Similar to the results on the CDD dataset and SYSU-CD dataset, it has better representation for the changed regions.

E. Ablation Experiments

In order to validate the efficiency of different modules of our presented network based on metric learning, we conducted ablation experiments on the proposed network on the CDD, SYSU-CD, and LEVIR-CD datasets, where we maintain the same parameters and details as the comparative experiments. We set up the ablation experiments in two parts, first, on the ablation experiments for our proposed joint learning network framework, and second, on the ablation experiments for the various proposed modules. Tables IV and V show the results of the two ablation experiments, respectively. In purpose of conducting an ablation study to validate the contribution of each constituent, we list the network structure of the ablation experiment. As shown in Table VI, where w/o and w/ represent "without" and "with," respectively.

1) Joint Framework Validation: As shown in Fig. 1, the designed network contains two main branches in the encoder stage, the backbone fusion subnetwork and the auxiliary difference subnetwork. The final model will get the feature vectors in both branches and update the network parameters in two branches, which will update the network parameters for the joint branch during the back propagation. Therefore, we reduce the complete SSANet into three network structures in the framework dimension, where the encoder of baseline uses a fusion subnetwork alone. The encoder of network1 uses the difference subnetwork alone. The encoder contains both the fusion subnetwork and the difference subnetwork as network2. At the same time, we replace the residual modules of all encoders with bottleneck modules to avoid the interference generated by the proposed modules, to prove the effectiveness of the joint learning framework.

Table IV presents the performance of baseline, network1, and network2 on the CDD, SYSU-CD, and LEVIR-CD datasets. First, benefiting from the advantages of the joint learning framework, the results of the network2 approach consistently outperform those of baseline and network1. Second, compared to the respective network1, baseline shows better results, but significantly lower performance than network2. The network1 approach exhibits higher accuracy rates but lower IoU and F1, which shows that the difference-based feature processing is prone to overfitting phenomenon during network learning. Combining the two cases, the joint learning framework can effectively aggregate the advantages of the two different branching networks and enhance the robustness of the network model, demonstrating that the joint learning network framework can consistently outperform the single branching network model.

2) Module Validation: We use the joint learning network based on the bottleneck residual (network2) as the base for module validation, and then gradually add our proposed MCA module, FDR module, and AW module to validate the effectiveness of each component. The MCA module is applied to the backbone fusion network phase of the encoder, the FDR module is applied to the auxiliary differential feature extraction stage of the encoder, and the AW module is applied to the decoder step.

a) Effectiveness of the MCA module: To analyze the performance of multicore channel correction attention, we add the MCA module in the fusion network stage of the encoder based on the network2, which is network3. As shown in Table V, compared with network2, the addition of the MCA module produces a significant increase in performance, with

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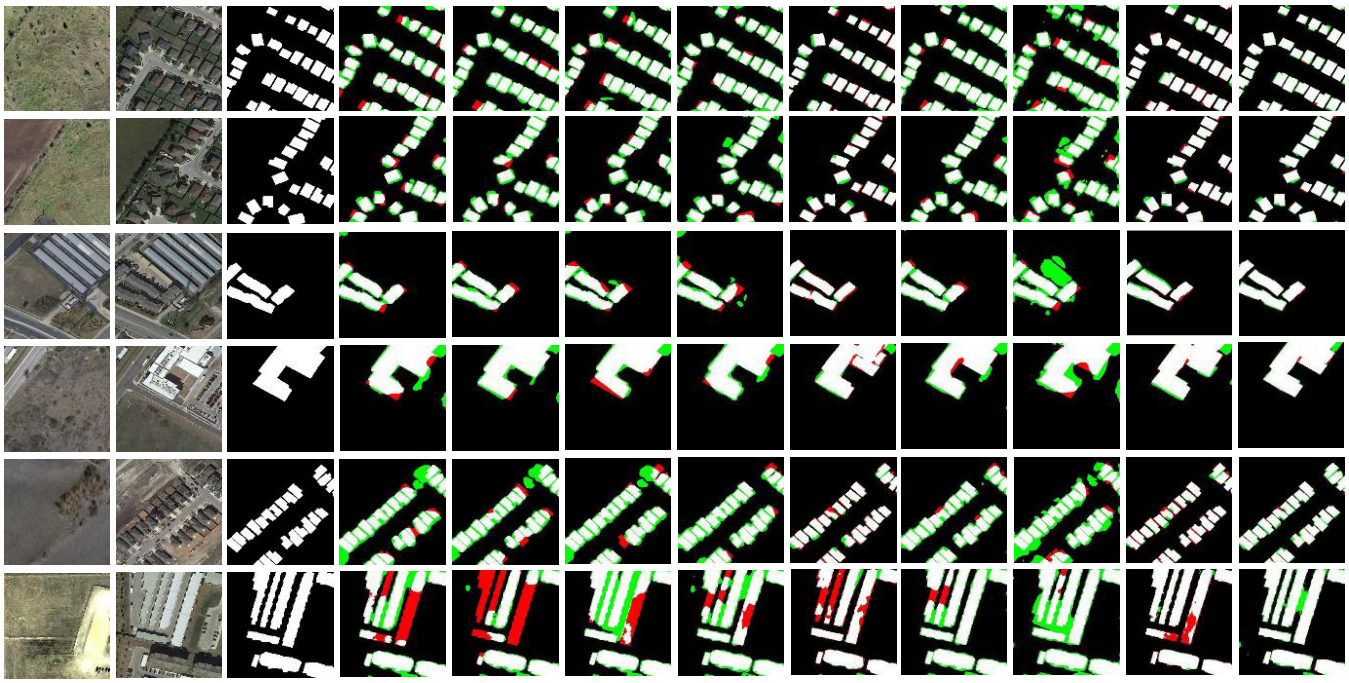


Fig. 12. Visual comparisons on the LEVIR-CD dataset: (a) image at time 1; (b) image at time 2; (c) ground truth; (d) FC-EF; (e) FC-Siam-diff; (f) FC-Siam-conc; (g) STANet; (h) BITNet; (i) SNUNet; (j) DSMANet; (k) DDCNN; and (l) proposed; colors: white for TP (i.e., "changed"), black for TN (i.e., "unchanged"), red for FP, and green for FN.

TABLE IV
ABLATION STUDY ON JOINT FRAMEWORK. ALL RESULTS ARE DESCRIBED AS %

	CDD				SYSU-CD				LEVIR-CD			
	Pre	Rec	F1	IoU	Pre	Rec	F1	IoU	Pre	Rec	F1	IoU
Baseline	97.28	91.30	94.19	89.03	82.53	67.60	74.32	59.13	89.88	84.45	87.08	77.13
Network1	92.80	92.48	92.64	86.30	80.77	63.15	70.88	54.90	86.91	66.97	75.65	60.83
Network2	97.31	93.41	95.32	91.07	84.83	69.70	76.53	61.98	91.07	86.27	88.61	79.55

TABLE V
ABLATION STUDY ON DIFFERENT MODULES. ALL RESULTS ARE DESCRIBED AS %

	CDD				SYSU-CD				LEVIR-CD			
	Pre	Rec	F1	IoU	Pre	Rec	F1	IoU	Pre	Rec	F1	IoU
Network2	97.31	93.41	95.32	91.07	84.83	69.70	76.53	61.98	91.07	86.27	88.61	79.55
Network3	98.23	94.63	96.40	93.06	83.76	74.86	79.06	65.38	90.35	89.61	89.98	81.80
Network4	99.05	95.20	97.09	94.36	81.63	78.76	80.16	66.90	90.15	90.68	90.42	82.51
SSANet	99.07	96.05	97.53	95.20	84.28	79.73	81.08	68.18	91.71	90.53	91.12	83.69

IoU increasing by 1.99%, 3.40%, and 2.25% on the CDD, SYSU-CD, and LEVIR-CD datasets, respectively. It indicates the importance of aggregation between multiscale spatial information and cross-channel attention. As expected from the experiments, the ablation experiments verify that the MCA module can achieve a good information interaction between local and global channel attention, thus producing finer multiscale spatial information.

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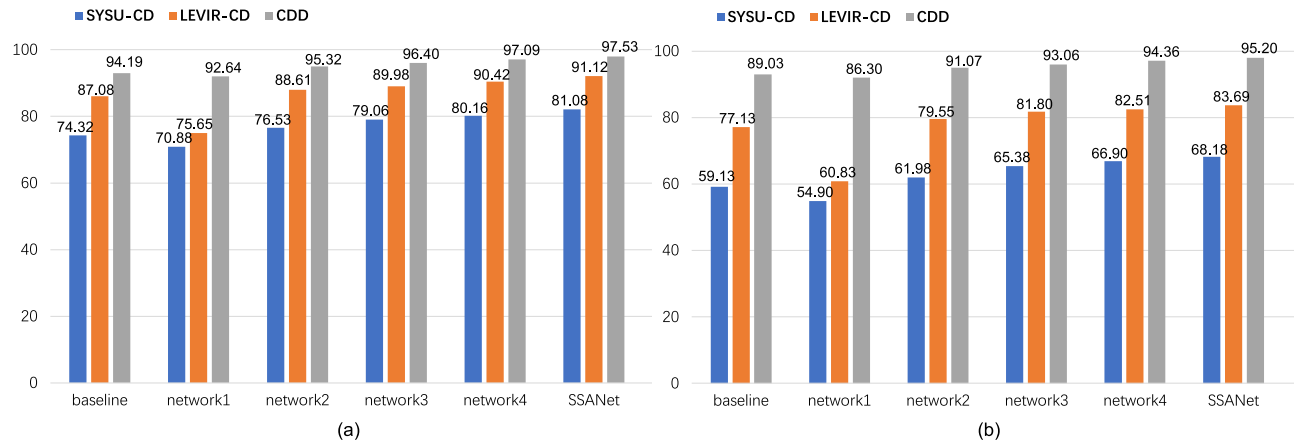


Fig. 13. Ablation experiments of: (a) F1 and (b) IoU.

b) Effectiveness of the FDR module: Since the FDR module is applied to the auxiliary difference net, the input of this net is the direct subtraction of the bitemporal image, and then used for feature learning. However, the difference map obtained from the direct difference has a large amount of noise, so the original intention of the FDR module is designed to suppress the effect of noise, so as to learn high-quality difference features. Therefore, we introduce the FDR module in the difference extraction network branch based on network3, which is network4. As shown in Table V, adding the FDR module for disparity feature extraction further increases the IoU of the network by 1.30%, 1.52%, and 0.71% on the CDD, SYSU-CD, and LEVIR-CD datasets, respectively. This shows that the FDR module can effectively reconstruct the regions closer to the real change in the difference map and effectively suppress the pseudo-change regions. The effectiveness of the FDR module is fully verified.

c) Effectiveness of the AW Module: To validate the effectiveness of our proposed AW module, based on network4, we introduce the AW module in the decoder stage, which is our proposed SSANet. From the results, it can be seen that IoU improved by 0.84%, 1.28%, and 1.18% compared with network4 on the CDD, SYSU-CD, and LEVIR-CD datasets, respectively. This shows that the AW module can flexibly extract fused feature information and difference feature information, which serves as a well-balanced effect, thus verifying the performance of AW module.

3) Metrics Visualization: To visualize the effect of each individual framework and module of the ablation experiment, we visualize F1 and IoU as shown in Fig. 13, where we can see that the network structure with difference subnetwork and encoder alone, i.e., network1, shows lower F1 and IoU on two datasets. This indicates that there is a significant noise impact with image differential as input alone. Network2 metrics demonstrate the effectiveness of the proposed framework. In the example of CDD dataset, network2 improves F1 and IoU compared to the baseline by 1.13% and 2.04%, respectively. Compared to network2, network3 has improved F1 and IoU by 1.08% and 1.99%, respectively. Network4 shows an increase of 0.69% and 1.30% compared to network3 for F1 and IoU, respectively. Finally, the F1 and IoU of the SSANet has improved by 0.44% and 0.84% compared to network4. In comparison to the baseline network, F1 and IoU has improved by 3.34% and 6.17%, respectively, which reflect the significant performance improvement of the network with the cumulative addition of each of the MCA, FDR, and AW modules.

TABLE VI
NETWORK STRUCTURE COMPOSITION OF THE ABLATION EXPERIMENT

Model	Network Structure
Baseline	Fusion net (w/o MCA) +Decoder (w/o AW)
Network1	Difference net (w/o FDR) +Decoder (w/o AW)
Network2	Fusion net (w/o MCA) +Difference net (w/o FDR) +Decoder (w/o AW)
Network3	Fusion net (w/ MCA) +Difference net (w/o FDR) +Decoder (w/o AW)
Network4	Fusion net (w/ MCA) +Difference net (w/ FDR) +Decoder (w/o AW)

V. DISCUSSION

A. Efficiency

Our original intention in designing the network is to achieve high accuracy while maintaining a small model size. To evaluate our model design, we evaluate the computational and space complexity for all compared networks in terms of the number of trainable parameters and floating-point operations per second (FLOPs), respectively. As shown in Table VII, the model parameters of the three UNet-based Siamese networks are smaller due to their adoption of fewer feature extraction layers. However, it leads to low performance and poor robustness when encountering complex detection tasks. After that, the SNUNet has a low number of trainable parameters and achieves better detection performance, but it

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TABLE VII
COMPARATIVE AND SPACE COMPLEXITY VERSUS PERFORMANCE ON CDD DATASET

Methods	Para. (M)	FLOPs (G)	F1(%)	IoU(%)
FC-EF	1.35	3.56	73.11	57.62
FC-Siam-diff	1.82	4.71	82.94	70.85
FC-Siam-conc	2.03	5.32	84.12	72.60
STANet	24.37	12.03	91.51	84.35
BITNet	22.87	26.31	94.45	89.50
SNUNet	13.21	54.82	94.49	89.57
DSAMNet	33.85	75.39	93.81	88.35
DDCNN	46.68	135.78	94.33	90.65
Ours	15.97	36.63	97.54	95.20

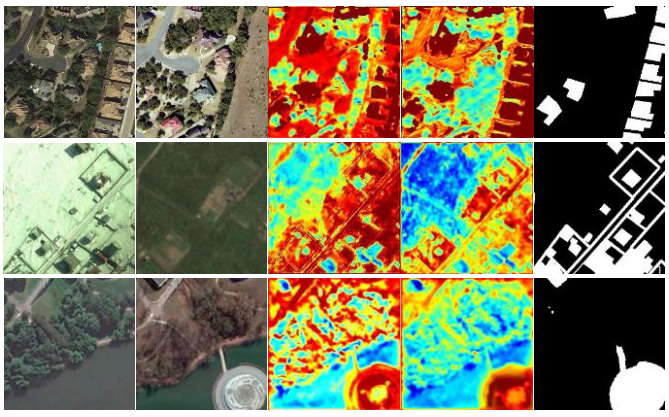


Fig. 14. Visualization of attention maps of the MCA module: (a) image at time 1; (b) image at time 2; (c) input of the MCA module; (d) output of the MCA module; and (e) label; colors: red denotes higher attention values and blue denotes lower values.

still has higher FLOPs. Then, we also compare with STANet, BITNet, SNUNet, DSAMNet, and DDCNN. Among them, it can be seen that the F1 and IoU of DDCNN are the highest, but it also has the highest space complexity. As shown in Table VII, our proposed method has much lower parameter numbers than STANet, BITNet, DSAMNet, and DDCNN, and the FLOPs are also smaller than SNUNet, DSAMNet, and DDCNN. Meanwhile, we also achieved the highest F1 and IoU in the comparison experiment. This indicates that our proposed method not only significantly reduces the number of parameters and model size, but also achieves the best detection performance.

B. Feature Visualization

Fig. 14 shows the visualization of the role of the MCA module using class activation mapping (Grad-CAM) method [54].

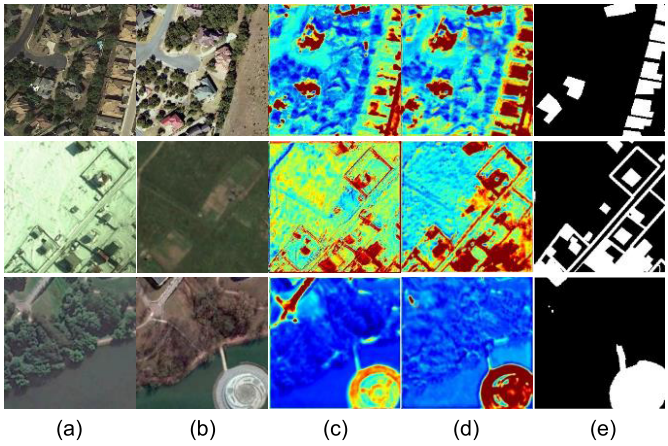


Fig. 15. Visualization of attention maps of the FDR module: (a) image at time 1; (b) image at time 2; (c) input of the FDR module; (d) output of the FDR module; and (e) label; colors: red denotes higher attention values and blue denotes lower values.

It can be observed that the spatial aggregation ability of the MCA module for features is more obvious, which can effectively aggregate multiscale contextual information to further extract more fine-grained HR features. However, it also exists the problem of focusing on unrelated pseudo-change regions, such as roads, houses, and trees. Therefore, we also visualize the features of the FDR module, as shown in Fig. 15. In contrast, it is observed that the FDR module is better to distinguish the area of variation. This is due to the FDR module's ability that effectively filters incorrect change information while retaining rich change details. It further demonstrates the effectiveness of our proposed module.

VI. CONCLUSION

In this article, we propose a joint learning network framework for CD task. The network architecture can realize joint learning of object features and difference features. Furthermore, to improve the feature extraction ability of the network, we introduce three modules in particular: MCA module, FDR module, and AW module. Specifically, MCA module can improve the capability of the network to extract object features. FDR module can well suppress noise variations and extract more accurate variance variations. AW module can effectively perform asymmetric aggregation of object features and variance features to obtain better variance results. The experimental results indicate that our proposed model has good generalization and robustness ability, and achieves the best results on the CDD, SYSU-CD, and LEVIR-CD datasets when compared to other state-of-the-art methods. In the future, we will explore more effective CD methods to meet the needs of complex scenarios.

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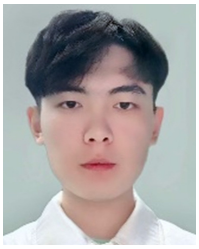
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